

4. Examination, Evaluation Procedure and Distribution of Marks

Each course evaluation covers two aspects:

- (i) Continuous Assessment (CA)
- (ii) Semester Final Examination

4.1 Evaluation of Theory Courses

4.1.1 All theory courses shall be evaluated out of 100 marks. The distribution of which is given below:

Sl		Items	Marks
1	CA	Attendance	5%
2		Assignment /Presentation	5%
3		Mid-Term Exam	20%
4	Semester Final Exam	Semester Final Exam	70%
	Total		100%

4.1.2 The Continuous Assessment of theory courses shall be conducted in the form of Attendance/ Class Participation, Assignment/Presentation, Quiz, and Mid-Term Exam. Class participation will be judged by in-class evaluation; attendance shall be recorded in every class.

4.1.3 The number of Assignment /Presentation of a course shall be at least 3 (Three). Average marks of the three assignments will be considered as final assignment marks.

4.1.4 After completing (40-50%) of the entire syllabus of a specific course, Mid-Term Exam shall be conducted and all students should confirm their participation in the mid-term examination.

4.1.5 Usually, dates of Mid-Term shall be fixed and announced accordingly by the respective course teacher or faculty of BOU. Duration of a Mid-Term Exam shall be 50-60 minutes, and it will preferably during the scheduled periods.

4.1.6 All Assignment/Presentation, Mid-Term Exam will be evaluated by the respective course teacher or faculty of BOU. At the end of the semester, the course teacher shall calculate total score of the Continuous Assessment, and shall submit the same to the Chairman of the respective Examination Committee before the semester final examination.

4.1.7 The concerned course teacher shall submit the class attendance register, evaluated assignments and answer scripts of mid-term to the Chairman of the respective Examination Committee.

4.1.8 Respective course teacher shall share (publish) the result of the Continuous Assessment to inform the students.

4.2 Evaluation of Lab Courses

4.2.1 All lab courses shall be evaluated out of 50 marks. The distribution of which is given below:

Sl		Items	Marks
1	CA	Attendance / Class Participation	10%
2		Assignment/Lab Report	10%
3		Experiment / Lab Based Project Works	30%
4	Semester	Problem Solving / Implementing Lab work	40%
5	Final Exam	Viva-voce	10%
	Total		100%

4.2.2 The Continuous Assessment of lab courses shall be conducted in the form of Attendance/ Class Participation, Assignment, Quiz, Lab Report, Experiment/Lab Based Project Work. Class participation will be judged by in-class evaluation; attendance will be recorded in every class.

- 4.2.3 The number of Assignment of a course shall be at least 3 (Three). Average marks of the three assignments will be considered as final assignment marks.
- 4.2.4 All Assignment, and Experiment/Lab Based Project Work will be evaluated by the respective course teacher or faculty of BOU. At the end of the semester, the respective course teacher shall calculate total score of the Continuous Assessment, and shall submit the same to the Chairman of the respective Examination Committee before starting of the semester final examination.
- 4.2.5 The course teacher shall also submit the class attendance register and other related documents to the Chairman of the respective Examination Committee.
- 4.2.6 Respective course teacher shall share the result of the Continuous Assessment to inform the students.
- 4.2.7 The internal examiner, who is the respective course teacher, and one external examiner, who possess expertise in the said course, jointly will conduct and evaluate the lab final exam. Questions will be set by the external and internal examiners during examination with the help of the Question paper supplied by the respective examination committee. Respective Examination Committee shall propose the name of the external examiner.
- 4.2.8 Viva-voce exam of Lab courses of a semester will be conducted by the Respective Examination Committee. If deemed necessary to the concerned Examination Committee, these sessions might be arranged online.
- 4.2.9 A student must attend the semester final exam and viva voce. In case of absence in any component, the student will get an F grade in that course.

4.3 Class Participation and Attendance Marks Distribution

The basis for awarding marks for class participation and attendance for each course either theoretical or lab will be as follows:

Percentage of Attendance	Percentage of Marks
90% and above	100%
85% to less than 90%	90%
80% to less than 85%	80%
75% to less than 80%	70%
70% to less than 75%	60%
65% to less than 70%	50%
60% to less than 65%	40%
Below 60%	0%

4.4. Duration of Semester Final Examinations

Semester Final Examination for each course either theory or lab shall be of 3 hours duration.

4.5 Answer Script (Semester Final) Evaluation

- 4.5.1 There shall be two examiners to examine the answer scripts of each course of the semester final examination. The 1st examiner shall receive the answer scripts from the Chairman of Examination Committee along with necessary supporting documents (question, top sheet, blank mark sheets, etc.) immediately after receiving appointment letter. On receipt of examined answer scripts from the 1st examiner, the Chairman of the Examination Committee shall send the answer scripts to the 2nd examiner along with supporting documents for evaluation.
- 4.5.2 Average of the marks given by two examiners shall be considered as final mark of each course of the semester final examination. If the variation of marks given by the first and second examiner

is more than 20% for any script, a '3rd (Third) Examiner' shall examine the scripts. In such case the average marks of the three examiners shall be taken as final mark for the respective course.

- 4.5.3 If 50% or more answer scripts of a particular course are considered to be evaluated by third examiner, all the answer scripts of that particular course shall go to the third examiner for further evaluation.
- 4.5.4 The examiners will give marks to every answer within the marks sheet and add them. If any student answers a greater number of questions that are asked for, then the examiner will scratch the last redundant answer(s). The examiners in no way should put or indicate any given marks in the script.

4.6 Project Evaluation

- 4.6.1 Capstone Project Design/thesis work on B. Sc. in CSE is carried out for 12 months in the 4th year 1st semester and 2nd semester. A group of 1-2 students works under one supervisor who will be a faculty of any recognized university having experience in relevant field. They take Capstone Project Design/thesis work in different fields of CSE available in the department.
- 4.6.2 The Examination Committee of the 4th year 2nd semester and the supervisor shall evaluate the project. The students individually present their work in front of the Examination Committee.
- 4.6.3 A total of 200 marks is allocated for the Capstone Project Design/thesis work, out of which 100 marks are allocated for project/thesis Défense and the remaining 100 marks are allocated for report writing along with other activities. The distribution of marks is as follows:

Part-I: Marks Distribution of Supervisor

Sl. No.	Parameters to be considered	Marks
1	Project /Thesis/Internee work	60
2	Project/Thesis/Internee report	40
Total		100

Part-II: Marks Distribution of Examination Committee

Sl. No.	Parameters to be considered	Marks
1	Presentation on the project by individual student*	40
2	Project Report*	20
3	Viva-voce*	40
Total		100

* Assessment rubrics are added on Appendix

5. Grading System and Calculation of SGPA and CGPA

5.1 Grading Scale

The Universal Grading System introduced by the University Grants Commission (UGC) of Bangladesh, will be followed which are given below. The total marks obtained by a student in each course will be converted into Letter Grade and Grade Point. The minimum passing grade in any course will be D, and students must be passed the semester final exam separately.

Numerical Grade	Letter Grade		Grade Point
80% and above	A +	(A Plus)	4.00
75% to less than 80%	A	(A Regular)	3.75
70% to less than 75%	A -	(A Minus)	3.50
65% to less than 70%	B +	(B Plus)	3.25
60% to less than 65%	B	(B Regular)	3.00
55% to less than 60%	B -	(B Minus)	2.75
50% to less than 55%	C +	(C Plus)	2.50
45% to less than 50%	C	(C Regular)	2.25
40% to less than 45%	D		2.00
less than 40%	F		0.00

Note: “F” is the failing grade.

5.2. Calculation of Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA)

Students’ performance will be evaluated based on Semester Grade Point Average (SGPA) in each semester and Cumulative Grade Point Average (CGPA) is the Average Grade Point of all semesters

Calculation of SGPA:

$$SGPA = \frac{\text{Total number of grade points earned in a Semester}}{\text{Total number of credit in all courses taken in a Semester}}$$

As an instance consider the following example:

Course Code	Credits	Grades	Grade Point Equivalent	Grade Points (Grade Point Equivalent * Credits)
0613-101	3	A+	4.00	4.00 * 3 = 12.00
0613-102	1	A	3.75	3.75 * 1 = 3.75
0311-101	3	A-	3.50	3.50 * 3 = 10.50
0231-101	3	B+	3.25	3.25 * 3 = 9.75
0222-102	3	B	3.00	3.00 * 3 = 9.00
0533-101	3	B-	2.75	2.75 * 3 = 8.25
0688-400	6	C+	2.50	2.50 * 6 = 15.00
Total	22			68.25

$$SGPA = \frac{68.25}{22} = 3.10$$

Calculation of CGPA:

$$CGPA = \frac{\text{Total number of grade points earned}}{\text{Total number of credit in all courses taken}}$$

6. Retake Examination of Course(s)

A student who fails in any course(s) in the Semester Final examinations or who registered for the course(s) but did not sit for the examination, the concerned course(s) will be considered as retake examination of course(s). The Continuous Assessment and the Semester Final examination will form the regular examination system, but the retake examination of course(s) will provide additional opportunities to improve the results of the students. The Retake examination will be guided by the following rules:

- 6.1 Student obtaining an 'F' grade in any course shall be required to improve the grade when offered in the subsequent semester(s), since achieving a passing grade ('D') by paying necessary re-examination fee(s).
- 6.2 Improvement of Grade: A student earning a grade of 'B-' (B minus) or lower grade in a course, s/he may also be allowed to Retake that course to improve the grade when offered in the subsequent semester(s). The following rules will apply in this regard:
 - 6.2.1 Improvement opportunity shall be once only for each course. In such case, learner shall have to pay twice of the normal re-examination fee for each course. However, improvement of a course shall not be allowed after the graduation.
 - 6.2.2 In the case of failure to improve his/her course grade at the improvement examination, the previous grade shall remain valid.
- 6.3 Both the cases stated in (6.1) & (6.2), the student shall not be allowed to repeat/re-appear Continuous Assessment for improvement of Continuous Assessment mark, if submitted once. For Retake examination, the marks of Continuous Assessment will remain same as previously obtained by the students. However, in case of any student (s) gets zero (0) mark, he/she will get a chance to reappear with prior permission of the Program Coordinator.